Practical Assignment 1

Problem Statement : Write menu based program-

a) Write a program to store roll numbers of student in array who attended a training program in random order. Write a function for searching whether a particular student attended a training program or not, using Linear search.

b) Write a program to store roll numbers of student array who attended training programs in sorted order. Write a function for searching whether a particular student attended a training program or not, using Binary search.

Input code:

```
#include <stdio.h>

void linearSearch(int a[], int n, int key);

void binarySearch(int a[], int n, int key);

void sortArray(int a[], int n);


int main()

{

  int a[50], n = 0, ch, i, key;


  do

  {


    printf("\nMenu:\n1. Linear search\n2. Binary search\n3. Exit\n");

    printf("Enter your choice: ");

    if (scanf("%d", &ch) != 1)

      break;


    if (ch == 1 || ch == 2)

    {
```

```
    printf("Enter no of students: ");
    scanf("%d", &n);

    printf("Enter roll numbers:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter rollno to be searched :");
    scanf("%d", &key);
}

switch (ch)
{
    case 1:
        linearSearch(a, n, key);
        break;

    case 2:

        sortArray(a, n);
        for (i = 0; i < n; i++) printf("%d ", a[i]);
        printf("\n");

        binarySearch(a, n, key);
        break;
```

```
            case 3:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
                break;
        }
    } while (ch != 3);

    return 0;
}

void linearSearch(int a[], int n, int key)
{
    int i, found = 0;

    for (i = 0; i < n; i++)
{
        if (a[i] == key)
      {
            printf("Element found at index %d (position %d)\n", i, i + 1);
            found = 1;
            break;
        }
    }
 if (!found)
    {
```

```
    printf("Element not found in the array!\n");
  }
}
void binarySearch(int a[], int n, int key)
{
  int low = 0, high = n - 1, mid;
  int found = 0;

  while (low <= high)
  {
    mid = low + (high - low) / 2;

    if (a[mid] == key)
    {
      printf("Element found at index %d (position %d)\n", mid, mid + 1);
      found = 1;
      break;
    } else if (a[mid] < key)
    {
      low = mid + 1;
    } else {
      high = mid - 1;
    }
  }

  if (!found) {
    printf("Element not found in the array!\n");
  }
```

```
}
void sortArray(int a[], int n)
{
   int i, j, temp;
   for (i = 0; i < n - 1; i++) {
      for (j = 0; j < n - i - 1; j++)
     {
         if (a[j] > a[j + 1]) {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
         }
      }
   }
}
```

Output:

```
Menu:
1. Linear search
2. Binary search
3. Exit
Enter your choice: 1
Enter no of students: 6
Enter roll numbers:
53 24 34 12 2 6
Enter rollno to be searched: 12
Element found at index 3 (position 4)
```

Menu:

1. Linear search

2. Binary search

3. Exit

Enter your choice: 2

Enter no of students: 5

Enter roll numbers:

8 64 75 46 51

Enter rollno to be searched :40

8 46 51 64 75

Element not found in the array!